

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2021-2022
FULL MARKS: 150

Math 4441: Probability and Statistics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 8 (eight) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses. Assume reasonable values for any missing data.

1. Suppose that 50% of the watermelons grown on a farm are classified as large, 30% as medium, and 20% as small. Suppose further that you buy five watermelons at random from this farm. Find the probability that:
 - a) At least two of them are large. 4
(CO2)
(PO1)
 - b) Two of them are large, two are medium, and one is small. 6
(CO2)
(PO1)
 - c) Three of them are large and two are medium. 6
(CO2)
(PO1)
 - d) Exactly two of them are medium if it is given that at least two of them are large. 4
(CO2)
(PO1)
2. The four-dimensional random vector X has the PDF given by

$$f_X(\mathbf{x}) = \begin{cases} 1, & 0 \leq x_i \leq 1, i = 1, \dots, 4; \\ 0, & \text{otherwise.} \end{cases}$$
 - a) Find the expected value vector, $E[X]$; 4
(CO2)
(PO1)
 - b) Find the correlation matrix, R_X ; 6
(CO2)
(PO1)
 - c) Find the covariance matrix, C_X . 6
(CO2)
(PO1)
3. A prisoner is trapped in a cell containing three doors. The first door leads to a tunnel that returns him to his cell after two days of travel. The second door leads to a tunnel that returns him to his cell after three days of travel. The third door leads immediately to freedom.
 - a) Assume that the prisoner will always select doors 1, 2, and 3 with probabilities 0.5, 0.3, and 0.2, respectively. Find the expected number of days until he reaches freedom. 6
(CO2)
(PO2)
 - b) Assuming that the prisoner is always equally likely to choose among those doors that he has not used, find the expected number of days until he reaches freedom. In this version, for instance, if the prisoner initially tries door 1, then when he returns to the cell, he will now select only from doors 2 and 3. 8
(CO2)
(PO2)

4. Random variables X and Y have a joint PMF described in Table 1:

Table 1: Joint PMF of random variables X and Y for Question 4.

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	$3/16$	$1/16$	0
$x = 0$	$1/6$	$1/6$	$1/6$
$x = 1$	0	$1/8$	$1/8$

- a) Find whether X and Y are independent or not.

5

(CO2)

(PO1)

- b) Now, perform the experiments sequentially from which X and Y are to be derived. First, find the value of the random variable X using the PMF of X . Then, find the value of Y from the conditional PMF of Y given X . The experiments require the PMF of X and the conditional PMF of Y given X are to be known. Find the PMFs and label the branch probabilities of the tree shown in Figure 1:

10

(CO2)

(PO1)

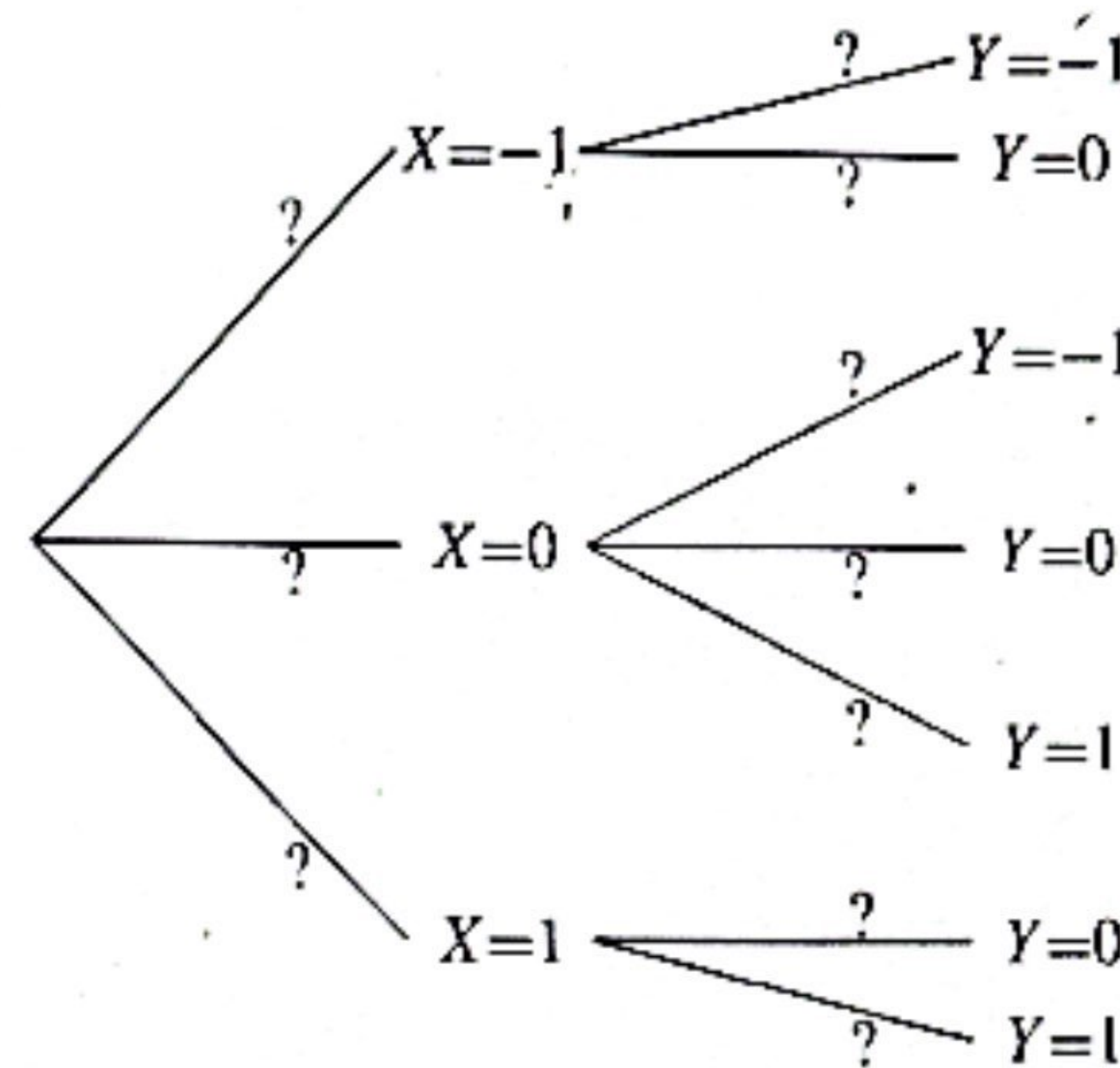


Figure 1: Tree for Question 4.b)

5. a) A fair coin is flipped until the first head occurs. Let X denote the number of flips required. Prove that the entropy of X , $H[X]$ is given by:

8

(CO2)

(PO1)

$$H[X] = -\log_2 p - \frac{1-p}{p} \log_2(1-p).$$

The following expressions may be useful:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \text{ and } \sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}$$

- b) Let random variables X and Y have the joint PMF shown in Table 2:

10

(CO2)

(PO1)

Table 2: Joint PMF of random variables X and Y for Question 5.b)

$P_{X,Y}(x,y)$	$y = 1$	$y = 2$	$y = 3$	$y = 4$
$x = 1$	$1/8$	$1/16$	$1/32$	$1/32$
$x = 2$	$1/16$	$1/8$	$1/32$	$1/32$
$x = 3$	$1/16$	$1/16$	$1/16$	$1/16$
$x = 4$	$1/4$	0	0	0

Find the mutual information between X and Y , $I(X;Y)$.

6. a) Given that random variable X has moment-generating function given by:

$$M_X(t) = \frac{1}{6}e^{-2t} + \frac{1}{3}e^{-t} + \frac{1}{4}e^t + \frac{1}{4}e^{2t}.$$

Find the PMF of X and calculate $P[|X| \leq 1]$.

7
(CO2)
(PO1)

- b) Let X be a random variable whose PDF is given by:

$$f_X(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Write down the MGF of X and use the MGF to calculate $E[X]$ and $Var[X]$.

8
(CO2)
(PO1)

- c) Let X be a discrete random variable with PGF:

$$G_X(s) = \frac{s}{5}(2 + 3s^2).$$

Find the distribution of X .

7
(CO2)
(PO1)

7. a) Suppose that $X_1, \dots, X_n$ form a random sample from a distribution for which the PDF, $f(x|\theta)$ is as follows:

$$f(x|\theta) = \begin{cases} \theta x^{\theta-1}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

10
(CO3)
(PO4)

Also, suppose that the value of θ is unknown ($\theta > 0$). Find the maximum likelihood estimator (MLE) of θ .

- b) In the June 1986 issue of *Consumer Reports*, some data on the calorie content of beef hot dogs is given. Here are the numbers of calories in 20 different hot dog brands:
186, 181, 176, 149, 184, 190, 158, 139, 175, 148, 152, 111, 141, 153, 190, 157, 131, 149, 135, 132.

10
(CO3)
(PO4)

Assume that these numbers are the observed values from a random sample of twenty independent normal random variables with mean μ and variance σ^2 , both unknown. Find a 90% confidence interval for the mean number of calories μ .

8. a) A fast-food establishment has been averaging about \$2000 of business per weekday. To see whether business is changing due to a deteriorating economy, management has decided to carefully study the figures for the next 8 days. Suppose the figures are
2050, 2212, 1880, 2121, 2205, 2018, 1980, 2188

3 + 7
(CO4)
(PO4)

- What are the null and the alternative hypotheses?
- Are the data significant enough, at the 5 percent level, to prove that a change has occurred?

- b) Two machines are used for filling plastic bottles with a net volume of 16.0 ounces. The fill volume can be assumed to be normal with standard deviation $\sigma_1 = 0.020$ and $\sigma_2 = 0.025$ ounces. A member of the quality engineering staff suspects that both machines fill to the same mean net volume, whether or not this volume is 16.0 ounces. A random sample of 10 bottles is taken from the output of each machine.

3 +
7 + 5
(CO4)
(PO4)

- What are the null and the alternative hypotheses?
- Do you think the engineer is correct? Use $\alpha = 0.05$.
- Find the p -value.

Machine 1		Machine 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Appendix

Necessary Equations/formulas

$$P_{X|A}(x) = \frac{P_X(x)}{P[A]}, x \in A$$

$$P_{XY}(x, y) = P_{X|Y}(x|y)P_Y(y)$$

$$P_{XY}(x, y) = P_X(x)P_Y(y), \text{ if } X \perp Y$$

$$P_X(x) = \sum_y P_{XY}(x, y) = \sum_y P_{X|Y}(x|y)P_Y(y)$$

$$E[X] = \sum_y E[X|Y = y]P_Y(y)$$

$$H[X] = - \sum_x P_X(x) \log P_X(x)$$

$$H[X, Y] = - \sum_y \sum_x P_{XY}(x, y) \log P_{XY}(x, y)$$

$$H[X, Y] = H[X] + H[Y|X]$$

$$I[X; Y] = H[X] - H[X|Y]$$

$$M_X(t) = E[e^{tX}] = \sum_x e^{tx} P_X(x) = \int_{-\infty}^{+\infty} e^{tx} f_X(x) dx$$

$$G_X(t) = E[z^X] = \sum_x z^x P_X(x)$$

Table 7.1 100(1 - α) Percent Confidence Intervals				
$X_1, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ $\bar{X} = \sum_{i=1}^n X_i/n$, $S = \sqrt{\sum_{i=1}^n (X_i - \bar{X})^2/(n-1)}$				
Assump- tion	Param- eter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$(-\infty, \bar{X} + z_{\alpha} \frac{\sigma}{\sqrt{n}})$	$(\bar{X} - z_{\alpha} \frac{\sigma}{\sqrt{n}}, \infty)$
σ^2 unknown	μ	$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$	$(-\infty, \bar{X} + t_{\alpha, n-1} \frac{S}{\sqrt{n}})$	$(\bar{X} - t_{\alpha, n-1} \frac{S}{\sqrt{n}}, \infty)$
μ unknown	σ^2	$\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right)$	$\left(0, \frac{(n-1)S^2}{\chi_{1-\alpha, n-1}^2} \right)$	$\left(\frac{(n-1)S^2}{\chi_{\alpha, n-1}^2}, \infty \right)$